

Topics in Quantitative Finance - Homework 2

Assigned: July 29, 2026. Due: **Friday, July 31, 2026** by 1PM.

Late homework **will not be accepted**.

Homework is to be done by each student individually. To receive full credit, you must email a completed copy of this Jupyter notebook to TAs at topics_in_qf@163.com by the due date and time. All codes must run correctly and solutions must be written up neatly in Markdown/LaTeX format. If you encounter problems with Jupyter notebook, please contact TA 李新宇 or 林文鑫.

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In the following, B_t denotes a standard Brownian motion.

Ito isometry

1. (20pts)

Find the variance of the following random variables by applying the Ito isometry

$$\mathbb{E} \left[\left(\int_a^b \varphi_s dB_s \right)^2 \right] = \int_a^b \mathbb{E} [\varphi_s^2] ds$$

- (a) $X = \int_0^t B_s^2 dB_s$. In this case, $\varphi_s = B_s^2$.
- (b) $X = \int_0^t \sqrt{s} B_s dB_s$. In this case, $\varphi_t = \sqrt{t} B_t$.

Solution 1(a).

Since the expectation of X is 0, and for $Y \sim N(0, t)$, $EY^4 = 3t^2$ (can be proved by Gaussian Integral easily), thus

$$\text{Var} X = \mathbb{E} \left[\left(\int_0^t B_s^2 dB_s \right)^2 \right] = \int_0^t \mathbb{E} [(B_s^2)^2] ds = \int_0^t \mathbb{E} [B_s^4] ds = \int_0^t 3s^2 ds =$$



Solution 1(b).

The expectation of X is 0.

$$\mathbb{E}[B_t^2] = t, \text{ so } \mathbb{E}[tB_t^2] = t\mathbb{E}[B_t^2] = t^2$$

$$\text{Var } X = \mathbb{E} \left[\left(\int_a^b \sqrt{t} B_t dB_t \right)^2 \right] = \int_a^b \mathbb{E} [(\sqrt{t} B_t)^2] dt = \int_a^b \mathbb{E} [t B_t^2] dt = \int_a^b t^2$$

Apply Ito's formula to evaluate stochastic integral

2. (20pts)

Apply Ito's formula to evaluate the following stochastic integrals.

- (a) $\int_0^t B_s e^{B_s} dB_s$
- (b) $\int_0^t \frac{1}{1+B_s^2} dB_s$

Solution 2(a).

Ito's Formula: $df(B_t) = f'(B_t) dB_t + \frac{1}{2} f''(B_t) dt$.

Since $\int x e^x dx = x e^x - e^x + C$,

$$\int_0^t B_s e^{B_s} dB_s = \int_0^t d(B_s e^{B_s} - e^{B_s}) - \frac{1}{2} \int_0^t (B_s e^{B_s} + e^{B_s}) ds = B_t e^{B_t} - e^{B_t} + 1$$



Solution 2(b).

$$\int_0^t \frac{1}{1+B_s^2} dB_s = \int_0^t d \arctan(B_s) - \int_0^t \frac{1}{2} \frac{-2B_s}{(1+B_s^2)^2} ds = \arctan(B_t) + \int_0^t \frac{B_s}{(1+B_s^2)^2} ds$$



Apply Ito's formula to solve SDE

3. (10pts)

Let X_t satisfy the SDE for Ornstein-Uhlenbeck process

$$dX_t = \lambda(m - X_t)dt + \sigma dB_t, \quad X_0 = x \geq 0,$$

where $\lambda > 0$, m and σ are constants.

Apply Ito's formula to $e^{\lambda t} X_t$ and show that the solution is given by

$$X_t = e^{-\lambda t} x + (1 - e^{-\lambda t}) m + \sigma \int_0^t e^{-\lambda(t-s)} dW_s.$$

Solution 3.

According to Ito's formula,

$$d(e^{\lambda t} X_t) = (e^{\lambda t}) dX_t + (\lambda e^{\lambda t} X_t) dt = (e^{\lambda t}) (\lambda(m - X_t)dt + \sigma dB_t) + (\lambda e^{\lambda t} X_t) dt \\ = (e^{\lambda t}) (\lambda m dt + \sigma dB_t)$$

Integrating both side from 0 to t,

$$e^{\lambda t} X_t - x = \int_0^t (e^{\lambda s}) (\lambda m ds + \sigma dB_s) = (e^{\lambda t} - 1)m + \sigma \int_0^t e^{\lambda s} dB_s$$

Multiply both side by $e^{-\lambda t}$, we get the solution:

$$X_t = e^{-\lambda t} x + (1 - e^{-\lambda t}) m + \sigma \int_0^t e^{-\lambda(t-s)} dB_s.$$

Substitute $B=W$ (both are standard brownian motion), we get the same equation as above.



Simulating the Black-Scholes model

4. (20pts)

With 100 steps within the time interval $[0, 1]$, simulate 20,000 sample paths for the Black-Scholes model

$$dX_t = rX_t dt + \sigma X_t dB_t$$

by the Euler-Maruyama scheme with $X_0 = 100$. The parameters are given by $r = 2\%$ and $\sigma = 25\%$.

- (a) Plot the histograms of X_t for $t = 0.4, 1$.
- (b) Evaluate the expectation $e^{-rT} \mathbb{E}[g(X_T)]$ by taking sample mean from using the simulated samples in (a), where $T = 1$ and g is given by

$$g(x) = \begin{cases} 5 & \text{if } x > 101, \\ -4 & \text{if } x \leq 101. \end{cases}$$

Plot the histogram of $g(X_1)$.

Note: In `python`, the function g can be coded as `5*(x>101) - 4*(x<=101)`.

In [1]: *# as always, import required modules and functions*

```
import numpy as np
from numpy import sqrt, log, exp
import matplotlib.pyplot as plt
import scipy.stats as ss
from scipy.stats import norm
import seaborn as sns
```

In [2]: *#(4a)*

```
# set random seed
np.random.seed(1609)

# Parameters of the Black-Scholes model
r, sigma = 0.02, 0.25
x0, T = 100, 1

# number of paths and number of time steps
n_sim, n_steps = 20_000, 100

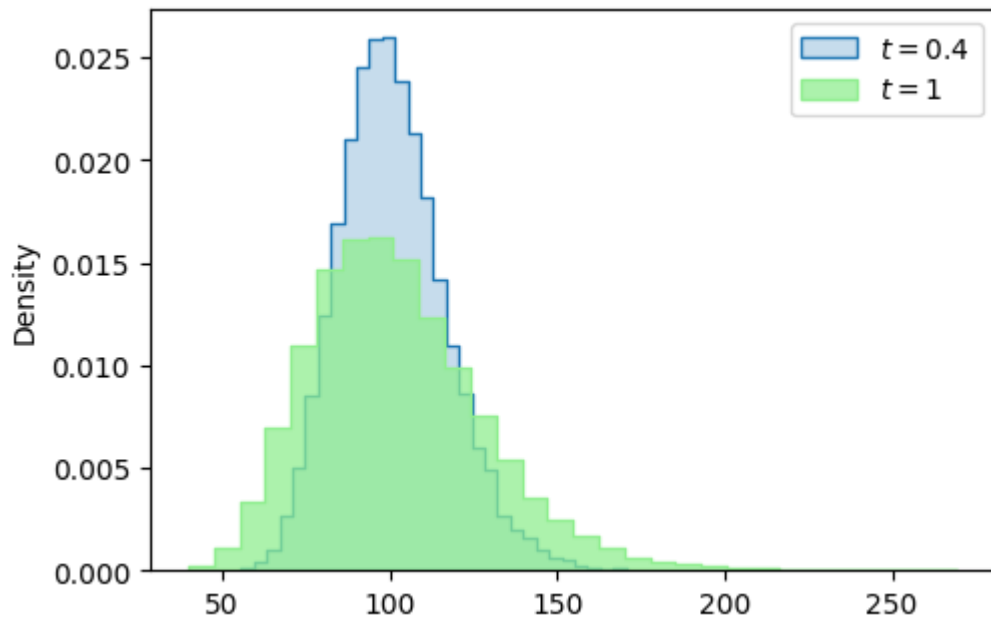
# initialize X
X = np.ones([n_sim, n_steps+1])*x0

# Euler-Maruyama scheme
dt = T/n_steps
for i in range(n_steps):
    db = norm.rvs(size=n_sim)
    db = db - db.mean()
    db = db/db.std()
    X[:,i+1] = X[:,i] + r*X[:,i]*dt + sigma*X[:,i]*np.sqrt(dt)*db # change here

# plot
plt.figure(figsize=(12, 8))
t = np.arange(0, T+dt, dt)

# histograms at different times
plt.subplot(2, 2, 1)
time = 0.4
sns.histplot(X[:,t==time], bins=30, stat='density', label=f'$t={time}$', element
sns.histplot(X[:, -1], bins=30, stat='density', label=f'$t={T}$', color='lightgre
plt.legend()
```

Out[2]: <matplotlib.legend.Legend at 0x14a3c9aee40>



```
In [3]: # 4b
g_X1 = 5 * (X[:, -1] > 101) - 4 * (X[:, -1] <= 101)
a = exp(-r * T) * np.mean(g_X1)

print("Discounted expectation:", a)
```

Discounted expectation: 0.22324024784561353

```
In [4]: # histogram of g(X1)
plt.hist(g_X1, bins=[-4.5, -3.5, 4.5, 5.5], density=True)

plt.xticks([-4, 5])
plt.xlabel("$g(X_1)$")
plt.ylabel("Density")
plt.title("Histogram of $g(X_1)$")
plt.show()
```

